# Iman Gholami

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## Research Interests

Approximation Algorithms, Network Design, Combinatorial Optimization, Algorithmic Game Theory, Metric Distortion, Formal Verification, AI for Mathematics

## Education

**University of Maryland, College Park** — MD, USA
*PhD Candidate in Computer Science* — *Jan 2023 – Present*
*MSc in Computer Science* — *May 2025*
- GPA: 3.97

**Sharif University of Technology** — Tehran, Iran
*BSc in Computer Engineering* — *May 2022*
- GPA: 3.57

## Experience

**Research Assistant** — Jan 2023 – Present
*University of Maryland* — *College Park, MD*
- Research on network design problems, including variations of classic connectivity problems, for better approximation factors than state-of-the-art algorithms.
- DARPA expMath team research towards verifiable mathematical proving through Artificial Intelligence (AI).
- Research on distortion problems, specifically classic problems with access to ordinal information instead of cardinal information.
- DARPA QuICC team research on Quantum-Inspired solvers for complex optimization problems.
- Academic mentoring of incoming PhD students and coordination of research teamwork.
- External reviewer for major conferences including FOCS, STOC, SODA, SPAA, and CDC.

**Teaching Assistant** — Jan 2023 – Present
*University of Maryland, College Park* — *MD, USA*
- Computational Genomics (Rob Patro, Spring 2026)
- Introduction to Computational Game Theory (MohammadTaghi Hajiaghayi, Fall 2025)
- Computer and Network Security (Michael Marsh, Fall 2024)
- Introduction to Computational Game Theory (MohammadTaghi Hajiaghayi, Fall 2023)
- Algorithms (Max Morowski & Clyde Kruskal, Spring 2023)

**Software Engineer Intern** — Jun 2026 – Aug 2026
*Intrinsic Technologies (Axiom)* — *New York, NY*
- Advancing formal verification of Zero-Knowledge Virtual Machines (ZK VMs) in Lean 4.
- Designed and implemented a Reinforcement Learning (RL) evaluation system to optimize Large Language Models (LLMs) and reasoning models for automated proof generation.

**Software Engineer Intern** — May 2025 – Aug 2025
*Axal (YC W25)* — *San Francisco, CA*
- Design and implement code analysis tools for information extraction using Artificial Intelligence (AI), with the goal of rewriting legacy code into modern, maintainable systems.

**Software Engineer and Team Leader** — Nov 2018 – Dec 2022
*Balad (Local Navigation and Maps App, 30M+ Installs)* — *Tehran, Iran*

**Teaching Assistant** — Sep 2019 – Jun 2022
*Sharif University of Technology* — *Tehran, Iran*
- Theory of Machines and Languages (Ali Movaghar, Fall 2021)
- Design Algorithm (Hamid Zarrabi-Zadeh, Fall 2021)
- Modern Information Retrieval (Mahdieh Soleymani Baghshah, Spring 2021)
- Discrete Structures (Hamid Zarrabi-Zadeh, Spring 2021)
- Design Algorithm (Mohammad Ali Abam, Fall 2019)

**Software Engineer Intern** — Jul 2018 – Oct 2018
*Bazaar (Local Application Marketplace, 100M+ Installs)* — *Tehran, Iran*

## Honors/Awards

**Best Paper Award FOCS 2025**
**Competitive University Fellowships (UMD Dean's Fellowship, MPower Early Scholars Award)**
**Silver Medal in International Olympiad in Informatics (IOI 2017)**
**2nd Place in ICPC NAS, Mid-Atlantic USA Region 2023**
**2nd Place in ACM/ICPC Asia Region, Tehran Site 2017**
**4th Place in ACM/ICPC Asia Region, Tehran Site 2020**
**Gold Medal in Iran National Olympiad in Informatics (INOI 2016)**
**Silver Medal in Iran National Olympiad in Informatics (INOI 2015)**
**Bronze Medal in National Olympiad of Computer Engineering 2021**

## Publications

**Nonuniform Buy-at-Bulk Network Design: Improved Approximation and LP Integrality Gap**
*arXiv preprint, 2026 | https://github.com/igholami/buy-at-bulk-lean*
**The Tight Competitive Ratio of Greedy for Online Steiner Forest**
*arXiv preprint, 2026 | https://github.com/igholami/online-steiner-forest-lean*
**Constructive Arboricity in Almost-Linear Time for Dense Graphs**
*arXiv preprint, 2026*
**Beyond the Library: An Agentic Framework for Autoformalizing Research Mathematics**
*arXiv preprint, 2026*
**Quiet Planting for $k$-SAT, Multiple Solutions of Arbitrary Geometry**
*Conference on Learning Theory (COLT 2026)*
**Bi-Criteria Metric Distortion**
*International Conference on Learning Representations (ICLR 2026)*
**Breaking a Long-Standing Barrier: 2-$\varepsilon$ Approximation for Steiner Forest**
*IEEE Symposium on Foundations of Computer Science (FOCS 2025)*
**2-Approximation for Prize-Collecting Steiner Forest**
*Journal of the ACM (JACM 2025), ACM-SIAM Symposium on Discrete Algorithms (SODA 2024)*
**Prize-Collecting Forest with Submodular Penalties: Improved Approximation**
*Integer Programming and Combinatorial Optimization (IPCO 2025)*
**Prize-Collecting Steiner Tree: A 1.79 Approximation**
*ACM Symposium on Theory of Computing (STOC 2024)*

## Skills

**Research & Leadership**: Technical Leadership, Teamwork, Communication, Conflict Resolution, Academic Mentoring, Research Strategy, Cross-functional Collaboration

**Theoretical Computer Science (TCS)**: Approximation Algorithms, Network Design, Algorithmic Game Theory, Combinatorial Optimization, Graph Theory, Data Structures, Problem Solving

**AI & Machine Learning**: Large Language Models (LLMs), Automated Theorem Proving, Generative AI, AI-driven Analysis, Applied ML

**Software Engineering**: Distributed Systems, System Architecture, Static Analysis, Large-Scale Backend Systems, High-Performance Computing, API Design

**Languages**: C++ (200K+ LOC), Python (200K+ LOC), Lean 4, Kotlin, TypeScript, Go, Java, Bash, SQL

**Frameworks & Libraries**: React, FastAPI, Django, Node.js, NumPy, Pandas, Scikit-learn

**Cloud, DB & Tools**: AWS, Docker, Kubernetes, PostgreSQL, MongoDB, Redis, Git, CI/CD, Linux, Nginx

## References

**Prof. Hajiaghayi, MohammadTaghi** hajiagha@umd.edu
*Minker Professor at Computer Science Department, University of Maryland, College Park*

**Dr. Chakraborty, Diptarka** diptarka@comp.nus.edu.sg
*Assistant Professor at School of Computing, National University of Singapore*

**Dr. Abam, Mohammad Ali** abam@sharif.edu
*Associate Professor at Computer Engineering Department, Sharif University of Technology*

**Dr. Zarrabi-Zadeh, Hamid** zarrabi@sharif.edu
*Associate Professor at Computer Engineering Department, Sharif University of Technology*